PROJECT REPORT

Airline Data Analysis

This project focuses on analysing airline operations data to extract insights related to **revenue, passenger trends, aircraft performance, and operational efficiency**.
This analysis helps stakeholders:

- Increase **profitability**
- Improve **resource allocation**
- Make **data-driven pricing decisions**
- Enhance **customer experience**

Tools Used-

- **Python (Pandas) for data analysis &**
- **Python (sqlite3) for the database connection**
- **Matplotlib & Seaborn for visualization**
- **Power BI for dashboard creation**

Business Problem

Identifies gaps between passenger demand and revenue generation, helping airlines optimize pricing, improve aircraft utilization, and maximize operational efficiency." The data was stored in a SQLite database, containing multiple relational tables.

- Which aircraft types generate the highest revenue?
- Is Business class significantly more profitable than Economy?
- Which fare conditions (Economy/Business) have the highest demand?
- Which aircraft has:
    - Highest average ticket value?
    - Highest passenger load?

Objectives-

1. Increase the Occupancy Rate: By increasing the occupancy rate, we can boost average profit earned per seat and mitigate the impact of the challenges we are facing.

2. Improve Pricing Strategy: Develop a comprehensive pricing strategy that takes into account market changes and customer preferences in order to attract and retain customers.

3.Enhance Customer Experience: Focus on providing seamless customer experience from booking to boarding.

Data Loading-

Handling a very large dataset consisting of crores of records, As the data was very large so it was in SQLite database. To efficiently manage this, a SQLite database connection was created and all the data files were loaded into the Data frame using pandas. Then perform all necessary analysis. And at last created a single consolidated summary table, which served as the base dataset for the visualization.

• Handled large dataset using SQLite

• Created connection to access database

• Used SQL to perform analysis

• Created a summary table

Data Cleaning-

Thers is no such cleaning. As the dataset was already clean, but I perform some basic data cleaning steps in the project.

- Verified if any null values exist in key columns like: Amount, Ticket no, Flight id, Fare conditions
- Checked for duplicate ticket entries to avoid overcounting passengers.
- Ensured correct data types
- Checked for Invalid Values

Metrices Created-

I created an occupancy rate metric to evaluate aircraft utilization and performed a what-if analysis by increasing occupancy by 10% to estimate its impact on revenue, helping identify potential growth opportunities without additional operational costs.

1. **Occupancy Rate** - Percentage of seats filled in an aircraft.
2. **10% Increased Occupancy Scenario** - Estimate potential growth in passengers
3. **10% Increased amount Scenario** - Estimated incremental revenue impact

Key Findings-

1. Plans which have more than 100 seats. So we can see there are only 6 plans which have more than 100 seats.

	aircraft_code	num_seats
0	319	116
1	320	140
2	321	170
3	733	130
4	763	222
5	773	402

2. number of tickets booked changed over time

Ticket bookings show a sharp initial growth followed by stabilization and a gradual rise toward peak demand, indicating strong seasonal trends and opportunities for revenue optimization. There is sudden drop at the end possible reason could be may be incomplete data of that day or for some reason demand drops.

3. Total amount earned changed over time?

Revenue shows a strong upward trend with an initial surge, followed by stable performance and a peak phase, indicating consistent demand and opportunities for revenue optimization. And then there is sharp decline at the last point.

4. Average charges for each aircraft with different fare conditions

Business Class Generates Highest Revenue-

- The Business class contributes the largest share of total revenue.
- Even with fewer passengers, higher ticket prices drive more earnings.

Economy Class Has Moderate Contribution-

- Economy class contributes a significant portion of revenue.
- Indicates:
    - High passenger volume
    - Lower ticket price compared to Business

Comfort class shows the lowest revenue contribution.
Possible reasons:

- Fewer seats available
- Lower demand compared to other classes
- Also only one Plan(773) have comfort seats available.

5**. For each aircraft, the total revenue per year and the average revenue per ticket.**

From the below table it is evident that, the aircraft with highest total revenue is **SU9** and from the bar plot in the previous section we can deduce that the price of business class and economy is the lowest in this aircraft. This can be a contributing factor for increase in revenue as more number of people might opt for this aircraft ticket owing to it's comparatively lesser cost.
The aircraft with the least total revenue is **CN1**, and the possible reason for this may be that it offers only economy class with the cheapest price per ticket and this may be due to poor conditions or less facilities and people may associate it with poor service and lower quality.

	aircraft_code	ticket_count	total_revenue	avg_per_ticket
0	319	52853	2706163100	51201
1	321	107129	1638164100	15291
2	733	86102	1426552100	16568
3	763	124774	4371277100	35033
4	773	144376	3431205500	23765
5	CN1	14672	96373800	6568
6	CR2	150122	1982760500	13207
7	SU9	365698	5114484700	13985

4. **The total turnover and occupancy rate could increase by giving all aircraft a 10% higher occupancy rate.**

	aircraft_code	booked_seats	num_seats	occupancy_rate	Inc occupancy rate
0	319	53.583181	116	0.461924	0.508116
1	321	88.809231	170	0.522407	0.574648
2	733	80.255462	130	0.617350	0.679085
3	763	113.937294	222	0.513231	0.564554
4	773	264.925806	402	0.659019	0.724921
5	CN1	6.004431	12	0.500369	0.550406
6	CR2	21.482847	50	0.429657	0.472623
7	SU9	56.812113	97	0.585692	0.644261

Dashboard

- The dashboard provides a comprehensive overview of key KPIs such as Total Revenue, Total Passengers, Total Flights, Avg. Occupancy rate.
- It includes bar charts to visualize total revenue for each Aircraft and each Fare Conditions.
- A donut chart to highlights the revenue for fare Class.
- A line chart to represent revenue trend over the time.
- A line chart to represent total bookings over the time.
- And some slicers like: Aircraft Code, Fare Conditions and Month Year.

Recommendations-

1. Focus on Premium (Business Class) Revenue

- Increase premium offerings (better services, loyalty perks)
- Target corporate and high-value customers

2. Improve Occupancy for Low-Performing Aircraft

- Reduce capacity on low-demand routes
- Offer discounts or promotions to fill seats

3. Dynamic Pricing Strategy

- Increase ticket prices during high-demand periods
- Offer early-bird discounts for low-demand periods

4. Seasonal Planning

- Increase flights during peak periods
- Allocate larger aircraft during high demand

Conclusion-

To summarize, analysing revenue data such as total revenue per year, average revenue per ticket, and average occupancy rate per aircraft is critical for airlines to maximize profitability. Airlines can find areas for improvement and modify their pricing to reflect the result of these indicators. A greater occupancy rate is one important feature that can enhance profitability by generating more revenue while minimizing operational costs associated with vacant seats.

The airline should revise the price for each aircraft as keeping the price too low or two high drives customer ticket purchasing behaviour. They should decide a reasonable price as per the facilities offered and the condition of the aircraft.